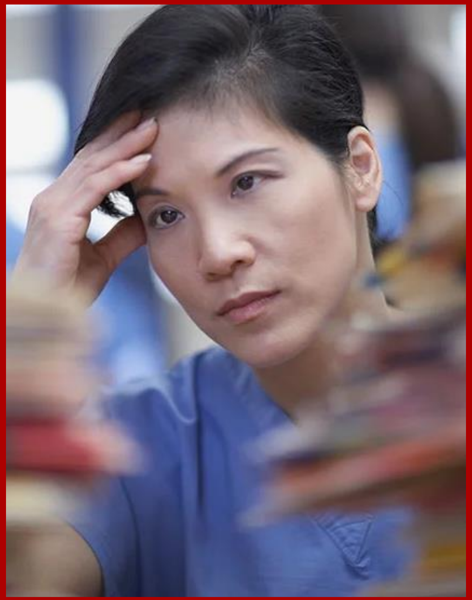
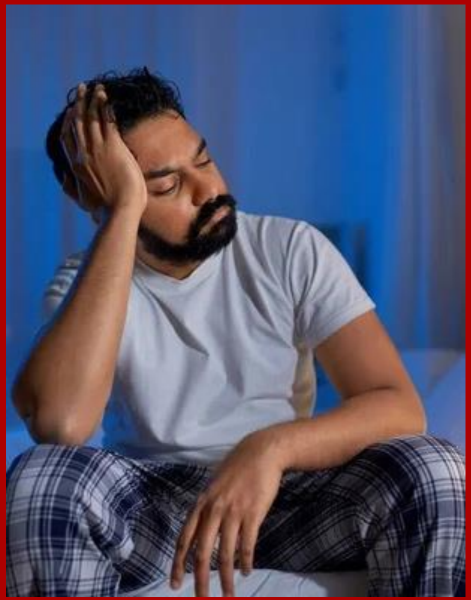


Treatment of Psychological Disorders 2: Biological Treatments

Lecture 20

Summer 2026

Dr. McPhee

By the end of class today, students should be able to…

- …Summarize **pharmaceutical interventions** for psychological disorders (**anti-psychotic, anti-anxiety,** and **anti-depressant medications**).
- …Describe other physiological interventions for mental illness, including **electroconvulsive therapy (ECT), transcranial magnetic stimulation (TMS),** and **deep brain stimulation (DBS)**.

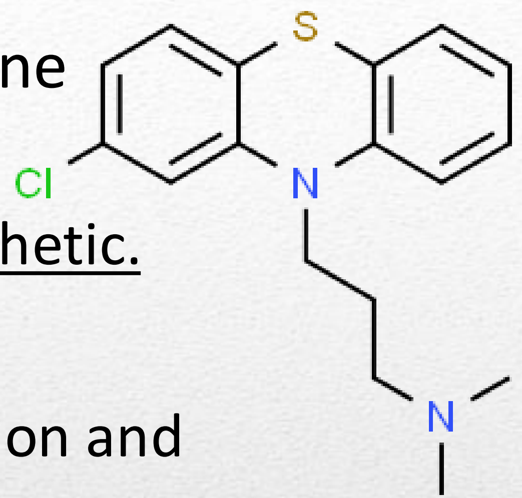
Daily Objectives for Today

- **Biological treatments** use therapeutic techniques to change an individual's physiology.
- They consist broadly of:
  - Medications (anti-psychotic, anti-anxiety, anti-depressants, herbal/natural products).
  - **Electroconvulsive Therapy** (ECT).
  - **Transcranial Magnetic Stimulation** (TMS).
  - **Psychosurgery** (destruction or repair of specific brain areas).
- We'll first focus on the most common of these, **medication,** specifically **psychopharmacotherapy**.

Examples of biological treatments.

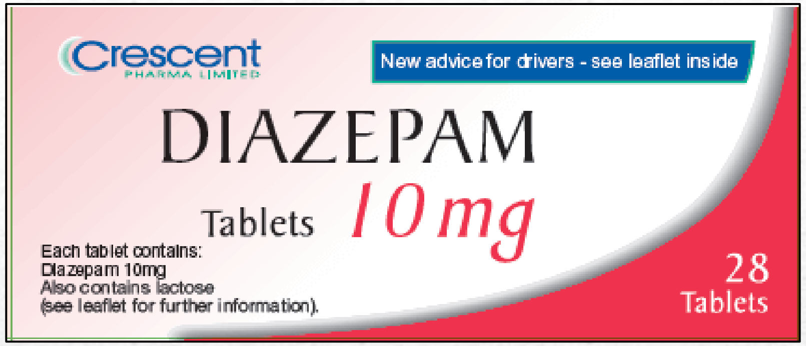
Biological Treatment

- The use of **medication** for psychological illness began with an accident.
- **Chlorpromazine** was first developed in France as an anti-histamine (allergy medication).
    - Also made patients sleepy. So, it started to be used as an anesthetic.
- Chlorpromazine blocks dopamine receptors**.**
    - **Dopamine** is a **neurotransmitter** associated with pleasure, motivation and motor control.
- Resulted in euphoric, calm patients instead of agitated patients.
- Introduction of these **anti-psychotic medications** changed the way that **schizophrenia** is treated.
    - But chlorpromazine and other **typical anti-psychotics** have significant side effects (e.g., **extrapyramidal symptoms** and **tardive dyskinesia**).



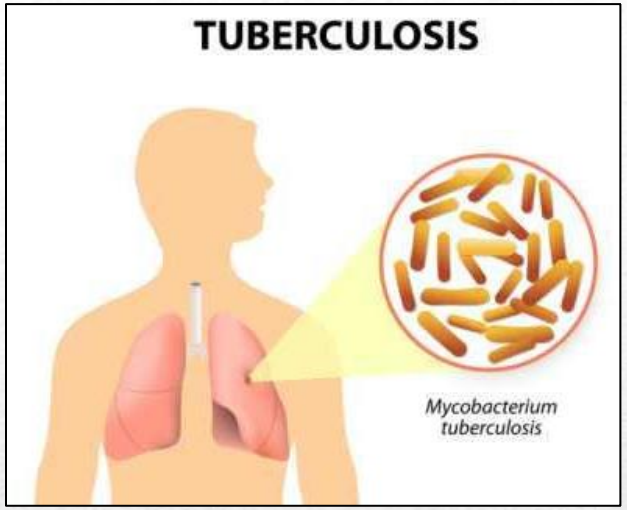
Anti-Psychotic Medication

- Since the discovery of anti-psychotics, other mental illnesses have come to be treated with medication.
- Anti-anxiety medications (**benzodiazepines**).
  - Facilitate GABA (gamma-aminobutyric acid) neurotransmitter activity → inhibit anxiety.
  - May be prescribed for people with panic disorders.
  - But:
    - **Drug Tolerance**
    - **Withdrawal Symptoms**
    - Side effects: drowsiness, poor coordination.
    - Highly addictive!



Diazepam is an example of a benzodiazepine.

Anti-Anxiety Medication



A patient with tuberculosis.

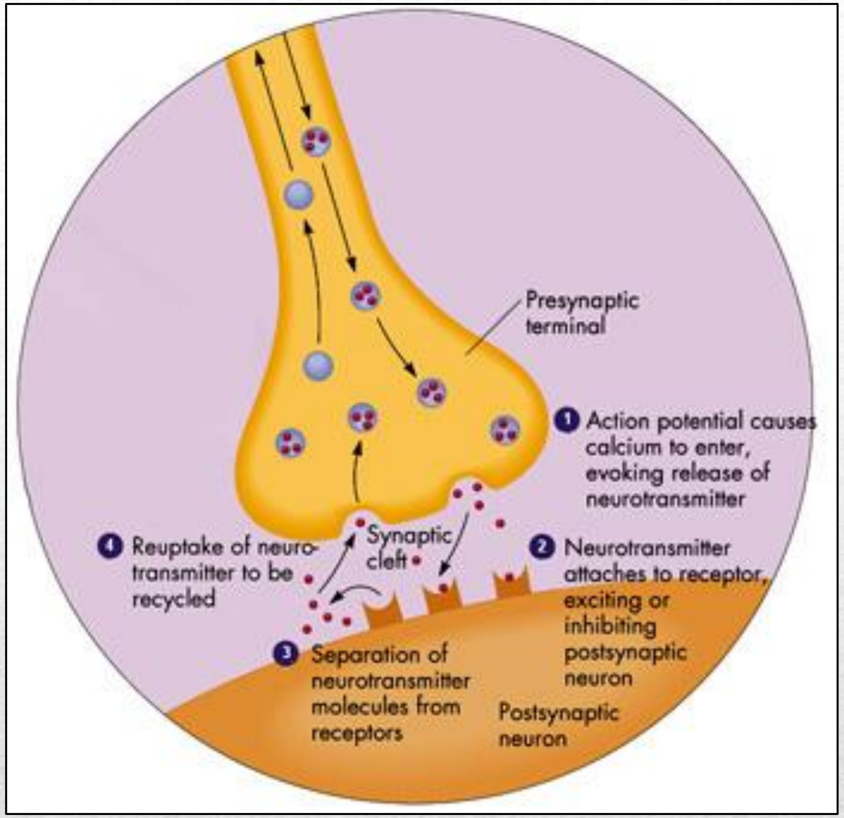
- **Anti-depressants** were also discovered somewhat accidentally.
  - They are medical treatments to treat depression.
- **Monoamine oxidase inhibitors**, used to treat tuberculosis in the 1950s, coincidentally elevated patients' moods.
  - **Monoamine oxidase** are enzymes that break down monoamines in the synapse, thereby disposing of serotonin, norepinephrine and dopamine.
  - If this enzyme is **inhibited**, then these mood enhancing neurotransmitters remain in the synapse for longer periods of time.

    → Prevented break-down of serotonin and dopamine.
  - Had intolerable side effects (dizziness, loss of sexual interest) BUT have dangerous (even lethal) food and drug interactions.

Anti-Depressant Medication

Anti-Depressant Medications

- Most anti-depressant medications today are **reuptake inhibitors.**
  - Prevent neurotransmitters from being taken back up.
  - Increase concentration of these neurotransmitters in the synaptic space.
  - Monoamine oxidase inhibitors prevent these neurotransmitters from being broken down in the system, thus increases the levels of them in the system.
  - These findings provided insight into the fact that neurotransmitters are related to mental health disorders.



Depiction of neurotransmitters being released into synaptic cleft.

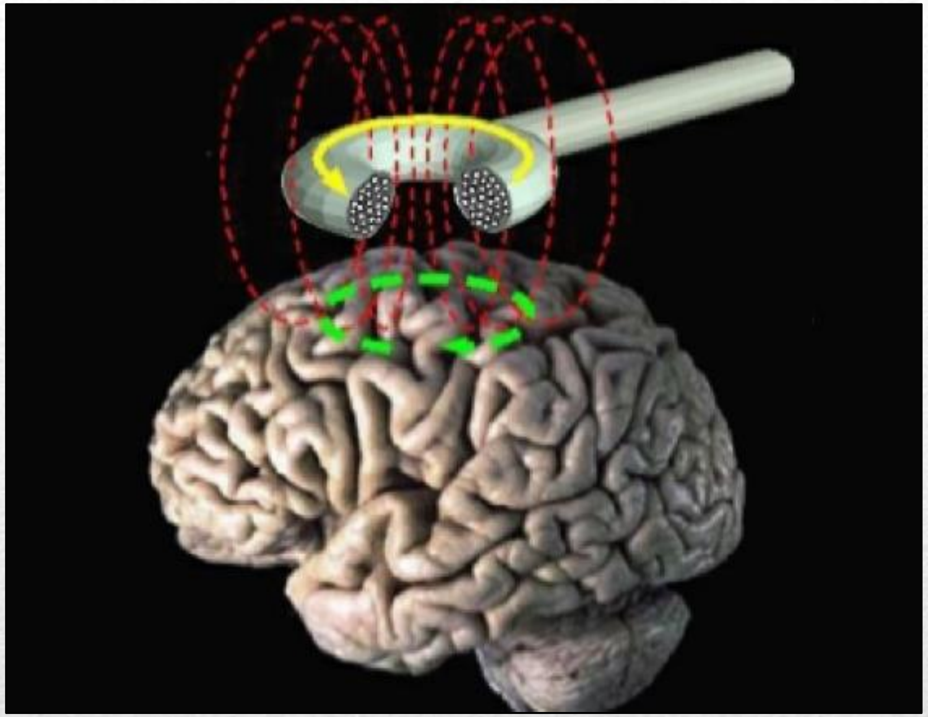
Anti-Depressant Medications

- It is believed that depression is a result of an imbalance of specific brain chemicals known collectively as monoamine neurotransmitters (includes **serotonin**, **norepinephrine** and **dopamine**).
- Each class of antidepressants acts on these neurotransmitters in slightly different ways.
- Reuptake inhibitors can work on many neurotransmitters or just one. Three major classes of antidepressants:
  1. Monoamine Oxidase Inhibitors
  2. **Serotonin only** (SSRIs; e.g., Cipralex).
     - SSRIs: Selective Serotonin Reuptake Inhibitors: makes more serotonin available at the synapse.
  3. **Serotonin and Norepinephrine** (Tricyclic antidepressants; e.g., Effexor).
     - Works on more than one type of neurotransmitter.

- **Electroconvulsive Therapy (ECT)**
  - Most commonly used in patients with severe major depression, bipolar disorder or schizophrenia that has not responded to other treatments.
  - ECT involves a brief electrical stimulation of the brain while the patient is under anesthesia. This induces controlled seizures in the brain.
  - Pros:
    - Highly effective and safe (in modern times).
    - Tends to work well for ~50-70% of people.
  - Cons:
    - Not permanent (needs to be repeated often).
    - There can be negative side effects like confusion and short-term memory loss (usually will go away after 7-14 days).

Other Physiological Interventions

- **Transcranial Magnetic Stimulation (TMS)**
  - A non-invasive procedure that uses small magnetic pulses (feels like light taps on the head) to stimulate nerve cells in the brain to improve symptoms of major depression.
  - Administered when other treatments haven't been effective.
  - The pulses pass through the skull and go about an inch into the brain where they can increase or decrease neuronal activity.
  - Considered to be "non-invasive".
  - Produces longer-lasting changes to brain chemistry.



Transcranial Magnetic Stimulation.

Other Physiological Interventions

- **Psychosurgery**: An operation in which psychological disorders are addressed using surgical procedures. Some forms remove or destroy small pieces of the brain thought to be causing mental dysfunction.
  - E.g., **Trepanation** was one of the earliest forms.
  - E.g., **Prefrontal Lobotomy**: developed to treat severe cases of psychosis. This procedure severs the connection between the prefrontal lobe and the rest of the brain.
  - E.g., **Deep Brain Stimulation (DBS):** a more modern form of psychosurgery.
- **Deep Brain Stimulation (DBS)**
  - An invasive surgical intervention where electrodes are embedded into the brain to produce repetitive brain stimulation.
  - The electrical current can help prevent signals that cause psychiatric symptoms.
  - It involves psychosurgery; associated with elevated risks. Typically used as a last resort for treatment-resistant cases of mental illness.

Other Physiological Interventions

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